

A critical reflection on refactoring the Web Witchcraft and Wizardry project

<https://witches-and-wizards.edmundmulligan.name>

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Abstract

This is a personal reflection on the design and implementation of a refactored version of a website. The new architecture is described, along with the navigation design, responsive and accessibility features, testing strategy, known issues and future improvements. Third party credits and resources are also acknowledged.

1 Introduction

Web Witchcraft and Wizardry (Mulligan, 2024) is a website for children to learn the basics of web development through fantasy-themed metaphors. The original version of the website was neither accessible (except by accident) nor responsive. This essay reflects on the design and implementation of a refactored version of the project, focusing on architecture, accessibility and code quality. Detailed decisions are documented as comments in the source code (available on the main branch at <https://github.com/Birkbeck2/puffin-web-development-website-edmundmulligan>), so this essay focuses on higher level design decisions and reflections.

For comparison the old and new landing pages can be seen in figures 1, 2 and 3.

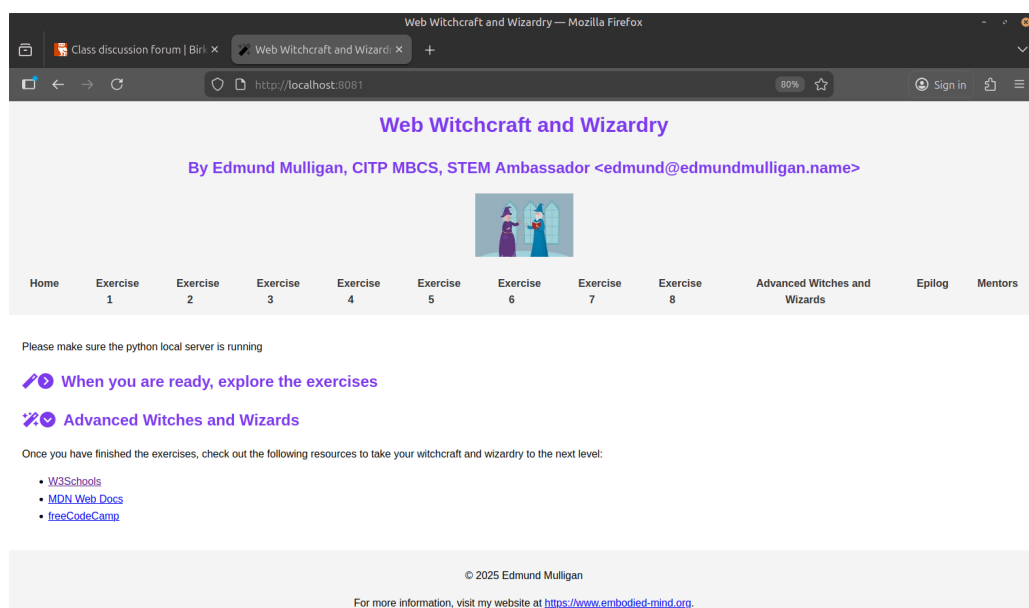


Figure 1: Old landing page



Figure 2: New landing page on a computer

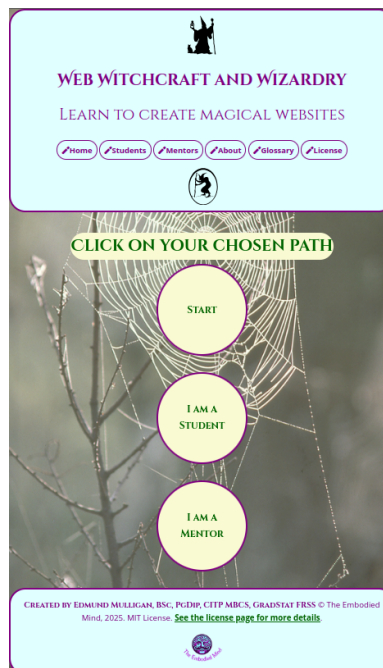


Figure 3: New landing page on a mobile device

The development process used was iterative and incremental (Beck and Andres, 2004), and while there are criticisms of this approach, particularly around its lack of documentation, endless changes and lack of scalability to large teams (see, for example, Boehm and Turner 2004), it was well suited to this small project. The advantage of being able to make quick changes and see immediate results (especially valuable when learning new techniques) was to some extent offset by the extra work required to frequently refactor code to maintain quality and consistency. The process is informally summarised in figure 4.

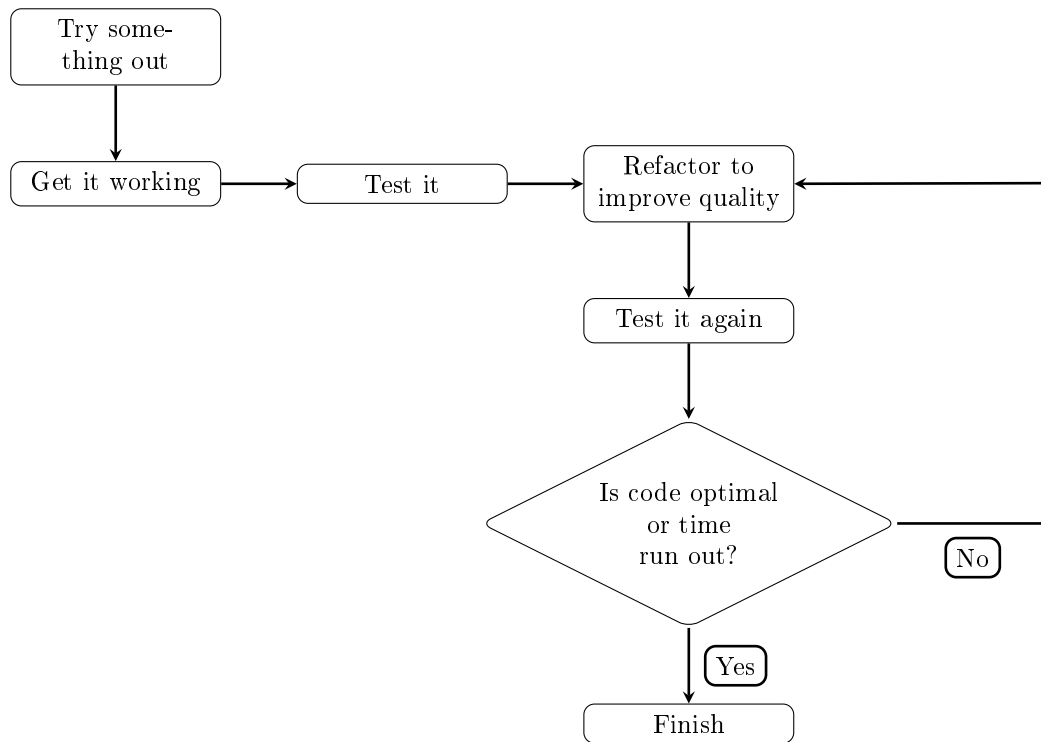


Figure 4: Iterative development process flowchart

2 Architecture

Even in a small project, software design principles are important to keep the code maintainable (Gamma et al., 1994). The three main principles applied in this project were separation of concerns (Parnas, 1972), DRY (Don't Repeat Yourself) (Hunt and Thomas, 1999), and self-documenting code (Martin, 2008). The first principle lead to the folder structure shown in figure 5. One advantage of this is that finding and fixing bugs is much easier in well structured code.

The second principle influenced the use of JavaScript to inject common elements into each page at runtime, rather than duplicating the code in each HTML file. This also inspired the use of CSS variables to define common colours and other styles in one place, rather than duplicating these values throughout the CSS files.

The third principle required meaningful names and comments and a consistent naming convention was used throughout.

A fourth (informal) principle is Keep the CSS as simple as possible and make use of cascading styles. This is something experienced web developers know and if I were to redo this project I would put much more effort into writing simple CSS first time rather than trying to refactor complex CSS later. The student.html page is a particular example where the CSS is more complex than necessary and needs more refactoring but I ran out of time to complete this.

```

root/
├── index.html
├── sitemap.xml
├── sitemap.xls
├── images/
│   ├── (site images)
│   └── fontawesome/
│       └── (svg fontawesome icons)
├── pages/
│   ├── about.html
│   ├── glossary-and-faq.html
│   ├── license-and-credits.html
│   └── students.html
├── scripts/
│   └── (injection JavaScript files)
└── styles/
    ├── index.css
    ├── main.css
    ├── media-queries.css
    ├── media-query-index.css
    ├── pages.css
    ├── noscript.css
    ├── students.css
    └── components/
        ├── buttons.css
        ├── header-and-footer.css
        ├── navigation.css
        └── titles.css

```

Figure 5: Directory structure of the project

3 Navigation

Navigation design was kept simple with a top navigation bar for site-wide navigation (figure 6) and an in-page navigation bar for navigating within all pages except the landing page (e.g. figure 7, although each page is slightly different). Incidentally, the wizard and witch in the site header are informally known to the development team as ‘Joe’ and ‘Helena’ respectively.



Figure 6: Site navigation bar structure

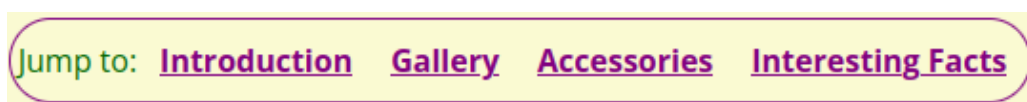


Figure 7: In-page navigation bar structure

4 A Responsive and Accessible Website

The main goal of the refactor was to make the website responsive and accessible. The primary techniques used to achieve responsiveness are media queries and flexible layouts (Marcotte, 2010). Accessibility features were mainly implemented by ensuring the code conformed to web standards, (Caldwell et al., 2024), but also by following best practices in the HTML markup, such as using semantic and landmark HTML elements (WHATWG, 2025),

4.1 Using and avoiding media queries

A point made in several YouTube videos (credited in section 7) is that while media queries are important, they should not be overused. The main layout should be flexible enough to adapt to a wide range of viewport sizes without the need for frequent breakpoints. Exactly which breakpoints to use is a matter of experience (which I lack) and judgement (which I possess in abundance), and I eventually settled on standardised breakpoints across all pages listed in table 1.

Viewport width	Layout	Impact
0px to 199px	Very small	Warn users but make best effort to display contents reasonably
200px to 799px	Mobile layout	Prefer vertical stacking
800px to 1199px	Desktop layout	Prefer horizontal layout
1200px and above	Large desktop layout	Take advantage of wider screen

Table 1: Viewport width breakpoints

In order to minimise the use of media queries, I used the `clamp()` CSS function to define fluid typography. A limitation of this is that if the middle value of `clamp()` is defined using a viewport width unit (vw), then the text will not scale if the user zooms their browser without changing the viewport window size. This contravenes WCAG 2.2 Success Criterion 1.4.4 (Caldwell et al., 2024) which requires text to be resizable up to 200% without loss of content or functionality. The workaround for this is to calculate the middle value of `clamp()` from a vw unit and a rem unit, so that zooming the browser window also changes the text size.

One downside of this approach is that it takes a lot of trial and error to find suitable values to pass to `clamp()` that work well across a wide range of viewport sizes and this took up much of my testing time. The process was made a lot easier when I discovered, at a late stage, an online tool to help calculate fluid typography values (Bece, 2025).

4.2 Flexbox vs Grid

One of the debates I encountered in many tutorials was whether to use CSS Flexbox (Atkins et al., 2018) or CSS Grid (Atkins et al., 2017) for layout. My conclusion was to use flexbox in narrower viewports as this could make vertical stacking of elements easier with `flex-direction: column`, and to prefer flexbox with `flex-direction: row` in wider viewports unless there was an obvious case for a grid such as the input form on the student page and the gallery of portraits. Due to my experimental approach to development, early versions of the website had several levels of nested grids and flexboxes which then had to be refactored to simpler layouts later. I also eventually realised that flexbox for vertical stacking is not always necessary as block elements naturally stack vertically. This is another area where more experience would have helped in making better design decisions earlier in the development process.

4.3 Units of measurement

Another way to achieve responsiveness is to use relative units of measurement rather than absolute units. Thus, I used `rem` units for specifying element sizes. I avoided using `px` units except in media queries where absolute measurements are required. I preferred `rem` units over `em` units as `rem`s are relative to the root font size, making them more predictable, whereas `em`s are relative to the font size of the parent element, which can (and did) lead to unexpected results.

4.4 Browser compatibility

Ensuring browser compatibility is an important part of web development, particularly when using newer CSS features that may not be supported in all browsers. I used caniuse.com (Can I Use, 2025) and only used HTML/CSS features that were supported by all major browsers (Chrome, Firefox, Safari and Edge). One feature I particularly wanted to try was `popover` to avoid having to use JavaScript `alert()`. I was mostly successful but could not get it to work with the form submit button (in `student.html`) so the Save Information button currently does nothing when clicked. That will be resolved with JavaScript in phase 2 of the project.

On a related note, all file paths were specified as absolute rather than relative. This is not best practice, but it avoided issues as I moved files around during development and refactoring. The website will break if it is not deployed into the root directory of a web server, but that is an acceptable limitation for now.

4.5 Other accessibility features

The accessibility testing tools sometimes flagged issues with ARIA (Diggs et al., 2023) compliance. I did not have time to fully investigate these issues, but I incorporated sufficient ARIA features to satisfy the testing tools. This certainly deserves further investigation in the future.

Choice of colours and contrast is very important, not only to make a site accessible but also visually appealing. This is another area where I have little design experience so I defer the aesthetic aspects of colour choice to next term's module, where that is covered more fully. For this phase of the project, I limited myself to ensuring that the colour choices were not too garish and that the choices passed the contrast tests in the accessibility tools. The purple and cyan colours were chosen as they (approximately) match the Embodied Mind logo used in the footer; the yellow and green were more arbitrary.

Testing without a mouse and with a screen reader ensured that that the `:focus` pseudo-class was styled appropriately to make it clear which element had focus and meaningful alt text was provided for all images. Using the web site in this way was not particularly pleasant or easy and emphasises how important it is to design for accessibility from the start to ensure that those who have no choice but to use assistive technologies have as good a user experience as can be achieved.

5 Testing

Testing software is not only a critical part of development (Myers et al. (2011), Mesbah et al. (2012)), but it is also difficult and time consuming. Many authorities (e.g. Boehm (1981), Jones (2008)) claim that testing can take at least 50% of total development time, and that concurs with my experience.

My approach to testing was to automate as much as possible using open source tools (see

Appendix B for results). The tests were designed to check for compliance with web standards (as a proxy for accessibility) and (given the age of the target users) text readability. Both static and dynamic testing tools (Myers et al., 2011) were used — static tools to check code validity, and dynamic tools to check accessibility of the rendered pages.

The tests were run frequently during development to catch issues early and ensure that the code remained compliant as it was refactored (regression testing). A github action was written to ensure the tests were run whenever code was pushed to the remote development and main branches in github. Testing tools were discovered using a google search and the most popular ones were chosen:

- Lighthouse (Google, 2025)
- Axe (Deque Systems, 2025)
- Pa11y (Pa11y Team, 2025)
- Wave (WebAIM, 2025)

A script to check reading age using the Flesch-Kincaid formula (Flesch, 1948) and Gunning Fog index (Gunning, 1952) was developed using the `text-readability` node.js module (Aashish Khanal, 2025). Code was checked for validity using node.js modules `html-validate` (html-validate, 2025), `stylelint` (Stylelint Team, 2025) and `eslint` (ESLint Team, 2025). The official W3C validator for CSS (W3C, 2025) was not used as this cannot currently validate code with CSS variables. Browser compatibility was tested using Playwright (Microsoft, 2025) to run tests in multiple browsers and I acknowledge, with thanks, the assistance I received to get this working from a senior tester colleague at TfL. This allowed automated testing of the latest versions of all three major browser engines (Chrome, Firefox and Safari). Playwright has potential for much more and could enable a test driven development (TDD) (Beck, 2003) approach in future phases of this project.

Implementing Wave testing was the most difficult as it requires a public URL to test and github pages are based on a private repository. Making the github repository public was considered and rejected as it could introduce unanticipated security risks so, after consultation with testing colleagues, `ngrok` (ngrok, 2025) was used to expose the localhost to allow Wave tests to complete.

There was insufficient time to fully evaluate the third party tools used and decide which was the best for this project, so a ‘black box’ approach was taken without investigating exactly what each tool was testing. This did have the benefit that gaps in one tool’s coverage could potentially be caught by one of the other tools, but it did incur a risk that the tools could change what they reported without notice. Indeed this occurred with Wave during the development of this project, when it stopped reporting that `<h1>` was in the `<header>` rather than `<main>` element. This brought it into line with the other tools, but decreases confidence in the reliability of the tools.

Automated testing is not a substitute for manual testing, and the site was also tested manually on a variety of devices, specifically

- A computer with a mouse and keyboard (running Linux Mint)
- A computer without using a mouse (to test keyboard navigation)
- A computer with a screen reader (Linux Orca)
- A touchscreen tablet device without mouse or keyboard (running Android)
- A mobile phone (running Android)

I did not have access to an iOS device so could not test on Apple hardware, nor did I have time to test on non-Linux operating systems. Testing was informal and ad hoc rather than systematic, and with more time, test scripts would be written to ensure consistent coverage across all devices. This would be more necessary if more than one developer were involved in the project to ensure everyone tested the same way.

I did not test the website with JavaScript disabled and I acknowledge this as a limitation of the current testing, driven by the available time. Testing thus far has mainly been verification rather than validation (Myers et al., 2011), (IEEE, 1990), but when actual lessons are implemented user and expert validation from a teacher will be needed to ensure the content is age appropriate and pedagogically sound. Security, and performance testing will also be needed when student data (tracking lesson progress) are collected.

6 Outstanding issues and future improvements

The project submitted for assessment is phase one of a larger project to refactor and improve the Web Witchcraft and Wizardry website. Some content was deliberately omitted from this phase as it will be developed in phase 2 using JavaScript. Appendix A lists some known issues and areas for improvement that could not be addressed within the time constraints of this phase.

7 Third party credits and resources

All code submitted for assessment is my own work. No code has been copied from any external sources, such as online tutorials or example code. I have, however, relied heavily on three YouTube channels, those of Kevin Powell (Powell, 2025), Daily CSS (Daily CSS, 2025), and Coding2GO (Coding2GO, 2025), for guidance on best practices and inspiration.

I made extensive use of Google Search, Google Scholar and Stack Overflow to research and troubleshoot specific issues I encountered.

The automation tools used for testing accessibility and performance (Lighthouse, Axe, pally, Wave) are all open source tools developed by third parties and the scripts used to run them were adapted from examples found in their respective documentations and in online resources. I was assisted by a colleague at my employer, Transport for London, in writing the Playwright tests used for testing browser compatibility.

The original artwork used in the website was created by Rachel Mulligan, a professional stained glass artist. I am grateful to her for allowing me to use her work in this project. The images used in this iteration are an early version and will be refined in future phases.

8 Conclusion

This has been a challenging and rewarding project. The challenge has not been coding the website, as I have 40 years of experience doing that, but in learning a lot of new material around modern CSS in a short space of time and applying that in a standards based development paradigm. The refactored Web Witchcraft and Wizardry website, while not yet complete, is a significant improvement on the original version in terms of accessibility, responsiveness and code quality. The use of automated testing has helped ensure that the code meets web standards and is accessible to the target audience. There are still some outstanding issues and areas for improvement that will be addressed in future phases of the project.

Word count

Word count: 2471 words (excluding references)

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Appendices

A Known Issues and Workarounds

This appendix contains known issues and workarounds.

Issue Id	Description	Symptoms	Work around
1	Responsive code does not work well at extreme viewport sizes.	Text overflows boxes or content does not fill screen properly.	Warning displayed when viewport is too small.
2	Font sizes are not optimal.	Text is too small or too large on some devices.	Further refinement of <code>clamp()</code> values is needed.
3	Bullets in famous witches and wizards on small screens don't work well.	Text margins too big so text wraps awkwardly.	Change from bullet list to definition list as that will format better on small screens.
4	student.css is a mess.	Doesn't conform to the standards in other css files.	More refactoring needed.
5	<code>popovertarget</code> does not work on submit buttons.	The popover does not appear when clicking the Save Information button.	Use JavaScript to show a popover or alert on form submission.
6	Website does not respect system light/dark mode preference.	Website always uses light mode even if system is in dark mode.	Use CSS media query <code>prefers-color-scheme</code> to adapt to system preference.
7	The artwork provided by Rachel Mulligan is in JPEG format with a white background.	The picture gallery looks ugly with white boxes around images.	Images need to be edited to have transparent backgrounds (PNG or SVG format).
8	Incomplete functionality.	Some features are not implemented yet.	Further development required to complete all features.
9	ARIA compliance is minimal.	Some elements lack full ARIA attributes.	Investigate ARIA, conduct accessibility audit and improvement code if needed.
10	Don't know how accessibility checkers work.	How do they handle checking at different viewports given the responsive design?	Need more investigation of these tools.

Table 2: Known issues and workarounds

B Automated Test Results

This appendix contains the complete test results from the automated test suite. All tests were executed on June 3, 2026 and demonstrate compliance with web standards, accessibility guidelines, and readability requirements.

Summary of Test Results

[Test Results] Test Results Summary
=====

[Code] Code Validation Results:

Files with issues: 0
HTML errors: 0
CSS errors: 0
Total errors: 0
Total warnings: 0

[Links] Broken Links Results:

Pages checked: 5
Total links: 125
Broken links: 0

=====

[OK] All tests passed (no critical/serious issues)

[Browser] Browser Compatibility Results:

Browsers tested: 3
Passed: 3
Failed: 0

[Axe] Axe Accessibility Results:

Total violations: 0
[Critical] Critical: 0
[Serious] Serious: 0
[Moderate] Moderate: 0
[Minor] Minor: 0

[Lighthouse] Lighthouse Accessibility Results:

Pages tested: 5
Average score: 100%
Pages with issues: 0
Total failed audits: 0

[Pa11y] Pa11y Accessibility Results:

Pages tested: 5
[X] Errors: 0
[Warning] Warnings: 0
Pages with errors: 0

[WAVE] WAVE Accessibility Results:

Pages tested: 5

[X] Errors: 0

[Warning] Alerts: 8

[Contrast] Contrast errors: 0

Pages with alerts: 5

[Reading] Reading Age Results:

Pages analyzed: 3

Total words: 2608

Average grade level: 9.5

[OK] All pages at high school level or below

Detailed Test Results

The following sections contain the complete JSON output from each test suite.

Code Validation Results

```
1 {  
  "files": [],  
  "summary": {  
    "htmlErrors": 0,  
    "htmlWarnings": 0,  
6    "cssErrors": 0,  
    "cssWarnings": 0,  
    "jsErrors": 0,  
    "jsWarnings": 0  
  }  
11 }
```

File Comments Results

File Header Comments Check Results
=====

=====

Summary

=====

Total files checked: 28
Files with issues: 0
 Missing header blocks: 0
 Missing required fields: 0

Broken Links Check Results

```
{
  "pages": [
    {
      "url": "index.html",
      "links": [],
      "brokenCount": 0,
      "totalCount": 18
    },
    {
      "url": "pages/students.html",
      "links": [],
      "brokenCount": 0,
      "totalCount": 16
    },
    {
      "url": "pages/glossary-and-faq.html",
      "links": [],
      "brokenCount": 0,
      "totalCount": 16
    },
    {
      "url": "pages/license-and-credits.html",
      "links": [],
      "brokenCount": 0,
      "totalCount": 28
    },
    {
      "url": "pages/about.html",
      "links": [],
      "brokenCount": 0,
      "totalCount": 47
    }
  ],
  "summary": {
    "totalLinks": 125,
    "brokenLinks": 0,
    "excludedLinks": 0
  }
}
```

Axe Accessibility Results

```
1 {  
  "violations": [],  
  "passes": [],  
  "incomplete": []  
}
```

Lighthouse Accessibility Results

```
{
  "pages": [
    {
      "url": "index.html",
      "score": 1,
      "failedAudits": []
    },
    {
      "url": "pages/students.html",
      "score": 1,
      "failedAudits": []
    },
    {
      "url": "pages/glossary-and-faq.html",
      "score": 1,
      "failedAudits": []
    },
    {
      "url": "pages/license-and-credits.html",
      "score": 1,
      "failedAudits": []
    },
    {
      "url": "pages/about.html",
      "score": 1,
      "failedAudits": []
    }
  ]
}
```

Pa11y Accessibility Results

```
1 {
  "pages": [
    {
      "url": "index.html",
      "documentTitle": "Web Witchcraft and Wizardry",
      "pageUrl": "http://localhost:8080/",
      "issues": []
    },
    {
      "url": "pages/students.html",
      "documentTitle": "Student Dashboard|Web Witchcraft and Wizardry",
      "pageUrl": "http://localhost:8080/pages/students",
      "issues": []
    },
    {
      "url": "pages/glossary-and-faq.html",
      "documentTitle": "Glossary and FAQ|Web Witchcraft and Wizardry",
      "pageUrl": "http://localhost:8080/pages/glossary-and-faq",
      "issues": []
    },
    {
      "url": "pages/license-and-credits.html",
      "documentTitle": "License and Credits|Web Witchcraft and Wizardry",
      "pageUrl": "http://localhost:8080/pages/license-and-credits",
      "issues": []
    },
    {
      "url": "pages/about.html",
      "documentTitle": "About|Web Witchcraft and Wizardry",
      "pageUrl": "http://localhost:8080/pages/about",
      "issues": []
    }
  ]
}
```

WAVE Accessibility Results

```
1 {
  "pages": [
    {
      "url": "index.html",
      "errors": 0,
      "alerts": 1,
      "features": 4,
      "structural": 11,
      "contrast": 0,
      "errorItems": [],
      "alertItems": [
        {
          "id": "noscript",
          "description": "Noscript element",
          "count": 1
        }
      ],
      "contrastItems": [],
      "waveUrl": "http://wave.webaim.org/report?url=https://crinkly-
        majorie-dramatic.ngrok-free.dev/index.html"
    },
    {
      "url": "pages/students.html",
      "errors": 0,
      "alerts": 2,
      "features": 17,
      "structural": 21,
      "contrast": 0,
      "errorItems": [],
      "alertItems": [
        {
          "id": "link_redundant",
          "description": "Redundant link",
          "count": 1
        },
        {
          "id": "noscript",
          "description": "Noscript element",
          "count": 1
        }
      ],
      "contrastItems": [],
      "waveUrl": "http://wave.webaim.org/report?url=https://crinkly-
        majorie-dramatic.ngrok-free.dev/pages/students.html"
    },
    {
      "url": "pages/glossary-and-faq.html",
      "errors": 0,
      "alerts": 2,
      "features": 4,
      "structural": 20,
```

```

51     "contrast": 0,
    "errorItems": [],
    "alertItems": [
      {
        "id": "link_redundant",
        "description": "Redundant link",
56         "count": 1
      },
      {
        "id": "noscript",
        "description": "Noscript element",
61         "count": 1
      }
    ],
    "contrastItems": [],
    "waveUrl": "http://wave.webaim.org/report?url=https://crinkly-
      majorie-dramatic.ngrok-free.dev/pages/glossary-and-faq.html"
66  },
  {
    "url": "pages/license-and-credits.html",
    "errors": 0,
    "alerts": 1,
71    "features": 4,
    "structural": 20,
    "contrast": 0,
    "errorItems": [],
    "alertItems": [
76      {
        "id": "noscript",
        "description": "Noscript element",
        "count": 1
      }
    ],
81    "contrastItems": [],
    "waveUrl": "http://wave.webaim.org/report?url=https://crinkly-
      majorie-dramatic.ngrok-free.dev/pages/license-and-credits.html
      "
  },
  {
86    "url": "pages/about.html",
    "errors": 0,
    "alerts": 2,
    "features": 33,
    "structural": 61,
91    "contrast": 0,
    "errorItems": [],
    "alertItems": [
      {
        "id": "heading_possible",
96        "description": "Possible heading",
        "count": 1
      },
      {

```

```
101     "id": "noscript",
        "description": "Noscript element",
        "count": 1
    }
],
106 "contrastItems": [],
    "waveUrl": "http://wave.webaim.org/report?url=https://crinkly-
        majorie-dramatic.ngrok-free.dev/pages/about.html"
}
]
```

Reading Age Analysis Results

```
1 {
  "pages": [
    {
      "file": "./pages/students.html",
      "wordCount": 500,
      "fleschReadingEase": 68.9,
      "fleschKincaidGrade": 8.4,
      "smogIndex": 11.5,
      "colemanLiauIndex": 9.2,
      "automatedReadabilityIndex": 10.3,
      "gunningFogIndex": 10.6,
      "daleChallScore": 7.1,
      "averageGradeLevel": 10,
      "textStandard": "8th and 9th grade",
      "difficultWords": 80,
      "averageSentenceLength": 19.2,
      "readingLevel": "Fairly Difficult (High School)",
      "readingAge": "17-18 years"
    },
    {
      "file": "./pages/glossary-and-faq.html",
      "wordCount": 81,
      "fleschReadingEase": 68.5,
      "fleschKincaidGrade": 6.5,
      "smogIndex": 10.3,
      "colemanLiauIndex": 8.9,
      "automatedReadabilityIndex": 7.1,
      "gunningFogIndex": 9.1,
      "daleChallScore": 7.8,
      "averageGradeLevel": 8.4,
      "textStandard": "8th and 9th grade",
      "difficultWords": 18,
      "averageSentenceLength": 11.3,
      "readingLevel": "Plain English (High School)",
      "readingAge": "15-16 years"
    },
    {
      "file": "./pages/about.html",
      "wordCount": 2027,
      "fleschReadingEase": 54.9,
      "fleschKincaidGrade": 9.6,
      "smogIndex": 11.7,
      "colemanLiauIndex": 9.8,
      "automatedReadabilityIndex": 9.7,
      "gunningFogIndex": 10.4,
      "daleChallScore": 7.3,
      "averageGradeLevel": 10.2,
      "textStandard": "9th and 10th grade",
      "difficultWords": 364,
      "averageSentenceLength": 16.3,
      "readingLevel": "Fairly Difficult (High School)",
```



```
    "readingAge": "17-18 years"  
  }  
]  
}
```

Cross-Browser Compatibility Results

```
{
  "browsers": [
    {
      "name": "chromium",
      "status": "passed",
      "error": null,
      "tests": [
        {
          "name": "Home page title",
          "status": "passed"
        },
        {
          "name": "Home navigation links",
          "status": "passed"
        },
        {
          "name": "Home JavaScript execution",
          "status": "passed"
        },
        {
          "name": "Home CSS Grid",
          "status": "passed"
        },
        {
          "name": "Home CSS Flexbox",
          "status": "passed"
        },
        {
          "name": "Home CSS Variables",
          "status": "passed"
        },
        {
          "name": "Home SVG support",
          "status": "passed"
        },
        {
          "name": "Home responsive viewport",
          "status": "passed"
        },
        {
          "name": "Home CSS calc()",
          "status": "passed"
        },
        {
          "name": "Home font rendering",
          "status": "passed"
        },
        {
          "name": "About page title",
          "status": "passed"
        }
      ]
    }
  ]
}
```

```

55      {
        "name": "About navigation links",
        "status": "passed"
      },
      {
        "name": "About JavaScript execution",
        "status": "passed"
      },
60      {
        "name": "About CSS Grid",
        "status": "passed"
      },
      {
65        "name": "About CSS Flexbox",
        "status": "passed"
      },
      {
70        "name": "About CSS Variables",
        "status": "passed"
      },
      {
        "name": "About SVG support",
        "status": "passed"
75      },
      {
        "name": "About responsive viewport",
        "status": "passed"
      },
80      {
        "name": "About CSS calc()",
        "status": "passed"
      },
      {
85        "name": "About font rendering",
        "status": "passed"
      },
      {
        "name": "Students page title",
90        "status": "passed"
      },
      {
        "name": "Students navigation links",
        "status": "passed"
95      },
      {
        "name": "Students JavaScript execution",
        "status": "passed"
      },
100     {
        "name": "Students CSS Grid",
        "status": "passed"
      },
      {

```

```

105     "name": "Students CSS Flexbox",
        "status": "passed"
    },
    {
110     "name": "Students CSS Variables",
        "status": "passed"
    },
    {
115     "name": "Students SVG support",
        "status": "passed"
    },
    {
        "name": "Students responsive viewport",
        "status": "passed"
    },
120    {
        "name": "Students CSS calc()",
        "status": "passed"
    },
    {
125     "name": "Students font rendering",
        "status": "passed"
    },
    {
130     "name": "Glossary & FAQ page title",
        "status": "passed"
    },
    {
        "name": "Glossary & FAQ navigation links",
        "status": "passed"
135    },
    {
        "name": "Glossary & FAQ JavaScript execution",
        "status": "passed"
    },
140    {
        "name": "Glossary & FAQ CSS Grid",
        "status": "passed"
    },
    {
145     "name": "Glossary & FAQ CSS Flexbox",
        "status": "passed"
    },
    {
        "name": "Glossary & FAQ CSS Variables",
150     "status": "passed"
    },
    {
        "name": "Glossary & FAQ SVG support",
        "status": "passed"
155    },
    {
        "name": "Glossary & FAQ responsive viewport",

```

```

    "status": "passed"
  },
160  {
    "name": "Glossary & FAQ CSS calc()",
    "status": "passed"
  },
165  {
    "name": "Glossary & FAQ font rendering",
    "status": "passed"
  },
170  {
    "name": "License & Credits page title",
    "status": "passed"
  },
175  {
    "name": "License & Credits navigation links",
    "status": "passed"
  },
180  {
    "name": "License & Credits JavaScript execution",
    "status": "passed"
  },
185  {
    "name": "License & Credits CSS Grid",
    "status": "passed"
  },
190  {
    "name": "License & Credits CSS Flexbox",
    "status": "passed"
  },
195  {
    "name": "License & Credits CSS Variables",
    "status": "passed"
  },
200  {
    "name": "License & Credits SVG support",
    "status": "passed"
  },
205  {
    "name": "License & Credits responsive viewport",
    "status": "passed"
  },
210  {
    "name": "License & Credits CSS calc()",
    "status": "passed"
  },
  {
    "name": "License & Credits font rendering",
    "status": "passed"
  }
]
},
{

```

```

"name": "firefox",
"status": "passed",
"error": null,
"tests": [
215   {
      "name": "Home page title",
      "status": "passed"
    },
    {
220     "name": "Home navigation links",
      "status": "passed"
    },
    {
      "name": "Home JavaScript execution",
225     "status": "passed"
    },
    {
      "name": "Home CSS Grid",
      "status": "passed"
230   },
    {
      "name": "Home CSS Flexbox",
      "status": "passed"
    },
235   {
      "name": "Home CSS Variables",
      "status": "passed"
    },
    {
240     "name": "Home SVG support",
      "status": "passed"
    },
    {
      "name": "Home responsive viewport",
245     "status": "passed"
    },
    {
      "name": "Home CSS calc()",
      "status": "passed"
250   },
    {
      "name": "Home font rendering",
      "status": "passed"
    },
255   {
      "name": "About page title",
      "status": "passed"
    },
    {
260     "name": "About navigation links",
      "status": "passed"
    },
    {

```

```

265     "name": "About JavaScript execution",
        "status": "passed"
    },
    {
        "name": "About CSS Grid",
        "status": "passed"
270    },
    {
        "name": "About CSS Flexbox",
        "status": "passed"
    },
275    {
        "name": "About CSS Variables",
        "status": "passed"
    },
    {
280        "name": "About SVG support",
        "status": "passed"
    },
    {
        "name": "About responsive viewport",
285        "status": "passed"
    },
    {
        "name": "About CSS calc()",
        "status": "passed"
290    },
    {
        "name": "About font rendering",
        "status": "passed"
    },
295    {
        "name": "Students page title",
        "status": "passed"
    },
    {
300        "name": "Students navigation links",
        "status": "passed"
    },
    {
        "name": "Students JavaScript execution",
305        "status": "passed"
    },
    {
        "name": "Students CSS Grid",
        "status": "passed"
310    },
    {
        "name": "Students CSS Flexbox",
        "status": "passed"
    },
315    {
        "name": "Students CSS Variables",

```

```

    "status": "passed"
  },
  {
320    "name": "Students SVG support",
    "status": "passed"
  },
  {
325    "name": "Students responsive viewport",
    "status": "passed"
  },
  {
    "name": "Students CSS calc()",
    "status": "passed"
330  },
  {
    "name": "Students font rendering",
    "status": "passed"
  },
335  {
    "name": "Glossary & FAQ page title",
    "status": "passed"
  },
  {
340    "name": "Glossary & FAQ navigation links",
    "status": "passed"
  },
  {
    "name": "Glossary & FAQ JavaScript execution",
345    "status": "passed"
  },
  {
    "name": "Glossary & FAQ CSS Grid",
    "status": "passed"
350  },
  {
    "name": "Glossary & FAQ CSS Flexbox",
    "status": "passed"
  },
355  {
    "name": "Glossary & FAQ CSS Variables",
    "status": "passed"
  },
  {
360    "name": "Glossary & FAQ SVG support",
    "status": "passed"
  },
  {
    "name": "Glossary & FAQ responsive viewport",
365    "status": "passed"
  },
  {
    "name": "Glossary & FAQ CSS calc()",
    "status": "passed"

```



```

370     },
    {
        "name": "Glossary & FAQ font rendering",
        "status": "passed"
    },
375     {
        "name": "License & Credits page title",
        "status": "passed"
    },
    {
380        "name": "License & Credits navigation links",
        "status": "passed"
    },
    {
        "name": "License & Credits JavaScript execution",
385        "status": "passed"
    },
    {
        "name": "License & Credits CSS Grid",
        "status": "passed"
390    },
    {
        "name": "License & Credits CSS Flexbox",
        "status": "passed"
    },
395     {
        "name": "License & Credits CSS Variables",
        "status": "passed"
    },
    {
400        "name": "License & Credits SVG support",
        "status": "passed"
    },
    {
        "name": "License & Credits responsive viewport",
405        "status": "passed"
    },
    {
        "name": "License & Credits CSS calc()",
        "status": "passed"
410    },
    {
        "name": "License & Credits font rendering",
        "status": "passed"
    }
415 ]
},
{
    "name": "webkit",
    "status": "passed",
420    "error": null,
    "tests": [
        {

```

```

    "name": "Home page title",
    "status": "passed"
425 },
    {
        "name": "Home navigation links",
        "status": "passed"
    },
430 {
        "name": "Home JavaScript execution",
        "status": "passed"
    },
    {
435         "name": "Home CSS Grid",
        "status": "passed"
    },
    {
440         "name": "Home CSS Flexbox",
        "status": "passed"
    },
    {
445         "name": "Home CSS Variables",
        "status": "passed"
    },
    {
450         "name": "Home SVG support",
        "status": "passed"
    },
    {
455         "name": "Home responsive viewport",
        "status": "passed"
    },
    {
460         "name": "Home CSS calc()",
        "status": "passed"
    },
    {
465         "name": "Home font rendering",
        "status": "passed"
    },
    {
470         "name": "About page title",
        "status": "passed"
    },
    {
        "name": "About navigation links",
        "status": "passed"
    },
    {
475         "name": "About JavaScript execution",
        "status": "passed"
    },
    {
        "name": "About CSS Grid",

```

```

    "status": "passed"
  },
  {
    "name": "About CSS Flexbox",
480    "status": "passed"
  },
  {
    "name": "About CSS Variables",
485    "status": "passed"
  },
  {
    "name": "About SVG support",
    "status": "passed"
  },
490  {
    "name": "About responsive viewport",
    "status": "passed"
  },
  {
495    "name": "About CSS calc()",
    "status": "passed"
  },
  {
    "name": "About font rendering",
500    "status": "passed"
  },
  {
    "name": "Students page title",
    "status": "passed"
505  },
  {
    "name": "Students navigation links",
    "status": "passed"
  },
510  {
    "name": "Students JavaScript execution",
    "status": "passed"
  },
  {
515    "name": "Students CSS Grid",
    "status": "passed"
  },
  {
    "name": "Students CSS Flexbox",
520    "status": "passed"
  },
  {
    "name": "Students CSS Variables",
    "status": "passed"
525  },
  {
    "name": "Students SVG support",
    "status": "passed"
  }

```

```

},
530 {
    "name": "Students responsive viewport",
    "status": "passed"
},
{
535 "name": "Students CSS calc()",
    "status": "passed"
},
{
    "name": "Students font rendering",
540 "status": "passed"
},
{
    "name": "Glossary & FAQ page title",
    "status": "passed"
545 },
{
    "name": "Glossary & FAQ navigation links",
    "status": "passed"
},
550 {
    "name": "Glossary & FAQ JavaScript execution",
    "status": "passed"
},
{
555 "name": "Glossary & FAQ CSS Grid",
    "status": "passed"
},
{
    "name": "Glossary & FAQ CSS Flexbox",
560 "status": "passed"
},
{
    "name": "Glossary & FAQ CSS Variables",
    "status": "passed"
565 },
{
    "name": "Glossary & FAQ SVG support",
    "status": "passed"
},
570 {
    "name": "Glossary & FAQ responsive viewport",
    "status": "passed"
},
{
575 "name": "Glossary & FAQ CSS calc()",
    "status": "passed"
},
{
    "name": "Glossary & FAQ font rendering",
580 "status": "passed"
},

```

```

    {
      "name": "License & Credits page title",
      "status": "passed"
    },
    {
      "name": "License & Credits navigation links",
      "status": "passed"
    },
    {
      "name": "License & Credits JavaScript execution",
      "status": "passed"
    },
    {
      "name": "License & Credits CSS Grid",
      "status": "passed"
    },
    {
      "name": "License & Credits CSS Flexbox",
      "status": "passed"
    },
    {
      "name": "License & Credits CSS Variables",
      "status": "passed"
    },
    {
      "name": "License & Credits SVG support",
      "status": "passed"
    },
    {
      "name": "License & Credits responsive viewport",
      "status": "passed"
    },
    {
      "name": "License & Credits CSS calc()",
      "status": "passed"
    },
    {
      "name": "License & Credits font rendering",
      "status": "passed"
    }
  ]
},
"summary": {
  "passed": 3,
  "failed": 0,
  "skipped": 0,
  "total": 3
}
}

```